

T3 Weinberg-coefficient RGE report

$T3 - d3 - d3 - F4$, $\alpha = 0$: $S_1 = (3, 0)$, $S_2 = (3, 1)$, $F = (4, \frac{1}{2})$

Matched EFT RGE	Success	Flavor RGE	Success
Neutrino mass	Success	Numerical running	Not run
Observables	Not run	Internal benchmark	Success

1 Conventions and scope

Below the common heavy-particle threshold, the active theory is SMEFT.

$$t = \ln \mu,$$

$$V(H) = \frac{\lambda_H}{2} (H^\dagger H)^2.$$

The flavor construction assumes three lepton generations, a diagonal heavy-fermion mass basis, and common scalar masses.

2 Matched Weinberg coefficient

$$C_5 = -\frac{2}{3} \left(-\frac{\sqrt{3}hM_F^3M_{S_2}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F^3M_{S_1}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} - \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} \right)$$

3 One-generation SMEFT check

The tensor calculation is required to reproduce the compact one-generation benchmark

$$16\pi^2\beta_{C_5} = C_5 \left[-3g_2^2 + 2\lambda_H + 6|y_u|^2 + 6|y_d|^2 - |y_e|^2 \right]. \quad (1)$$

The RGE stage stops with an error if the calculated expression does not reduce to this benchmark.

4 Three-generation flavor result

The matched one-generation result is factorized as

$$C_5^{(1)} = F_{\text{loop}} \bar{y}_1 \bar{y}_2,$$

and lifted to the symmetric flavor matrix

$$(C_5)_{pq} = \frac{1}{2} \sum_r F_r \left[y_{1,pr}^* y_{2,qr}^* + y_{2,pr}^* y_{1,qr}^* \right]. \quad (2)$$

Writing this matrix as $K \equiv C_5$, the implemented RGE is

$$16\pi^2 \frac{dK}{d\ln\mu} = (2\lambda_H - 3g_2^2 + 2T)K - \frac{3}{2} [Y_e Y_e^\dagger K + K(Y_e Y_e^\dagger)^T], \quad (3)$$

$$T = \text{Tr} \left(Y_e Y_e^\dagger + 3Y_u Y_u^\dagger + 3Y_d Y_d^\dagger \right). \quad (4)$$

The fully expanded symbolic matrix is retained as a data file rather than printed here, because it is much less readable than the matrix equation above.

5 Neutrino-mass conversion

The current symbolic and numerical implementation uses

$$m_\nu = -v^2 C_5. \quad (5)$$

Normalization check required. Earlier function documentation stated $m_\nu = -(v^2/2)C_5$, whereas the executed code uses $m_\nu = -v^2 C_5$. The extracted operator normalization and this conversion must be checked together before final phenomenology.

6 Numerical running and observables

Numerical running was not requested. Supply -numerical CONFIG.json to include this section.

7 Saved data

Result	File
Matched coefficient	data/c5_coefficient.txt
Rendered matched coefficient	c5_coefficient.pdf
One-generation beta function	data/c5_beta.txt
Full flavor beta matrix	data/c5_flavor_beta_matrix.txt
Symbolic neutrino-mass matrix	data/neutrino_mass_matrix.txt